

NEWTON 5 ENE, MS, USA

WMO: 747710

Lat: 32.3378N

Lon: 89.0703W

Elev: 114

StdP: 99.96

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 63831

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-3.7	-13.6	1.2	-0.5	-10.0	1.6	-0.4	6.3	8.1	5.6	7.1	0.8	N/A	0.325

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	34.4	24.2	33.4	24.3	32.4	24.2	26.3	31.3	25.8	30.7	25.4	30.2	2.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.5	28.5	24.6	19.8	28.0	24.1	19.2	27.5	82.4	31.4	80.1	30.8	78.2	30.2	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 4.9	4.1	3.6	-9.3	35.7	2.4	1.5	-11.0	36.7	-12.4	37.6	-13.7	38.4	-15.4	39.4		
(5)			-10.1	27.7	2.9	0.6	-12.2	28.1	-13.9	28.5	-15.5	28.9	-17.6	29.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	7.4	9.5	13.5	17.1	21.5	25.3	26.3	26.2	23.7	17.9	11.9	8.9		
	DBStd	7.96	5.81	5.55	4.94	3.92	3.22	1.85	1.57	1.90	2.77	4.62	4.88	5.45		
	HDD10.0	345	123	71	23	2	0	0	0	0	0	2	34	89		
	HDD18.3	1395	341	251	161	67	12	0	0	0	2	66	200	296		
	CDD10.0	3077	42	58	132	216	357	458	504	502	412	247	92	57		
	CDD18.3	1085	1	5	12	31	111	208	246	244	163	52	7	4		
	CDH23.3	9728	2	22	98	303	993	1951	2208	2171	1431	494	47	9		
	CDH26.7	3964	0	1	7	41	328	873	1000	996	581	134	4	0		
(14) Wind	WSAvg	1.4	1.7	1.8	1.8	1.6	1.3	1.1	1.0	1.0	1.1	1.2	1.4	1.6		
(15) Precipitation	PrecAvg	1450	152	139	147	138	110	111	128	118	88	88	105	134		
	PrecMax	1783	308	312	233	369	246	256	246	252	244	249	273	213		
	PrecMin	923	70	32	21	34	28	18	62	46	10	0	19	36		
	PrecStd	227	63	71	53	81	54	56	49	54	54	65	63	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.4	26.1	27.7	29.1	32.5	35.3	35.2	36.3	34.5	31.9	27.3	24.8		
		MCWB	17.1	19.2	18.5	21.0	22.6	24.0	24.4	25.3	22.6	19.7	19.4	20.6		
	2%	DB	21.3	23.3	26.0	27.9	31.1	33.9	33.9	34.4	32.9	29.8	24.6	22.5		
		MCWB	17.3	18.2	18.5	20.3	22.5	24.4	24.9	24.6	23.0	20.9	18.6	19.5		
	5%	DB	19.4	21.1	24.2	26.7	30.0	32.5	32.8	33.0	31.7	28.2	22.6	20.3		
		MCWB	15.9	17.3	17.5	19.3	21.8	24.4	24.8	24.8	23.2	20.0	17.7	17.8		
	10%	DB	17.3	19.1	22.1	25.2	28.8	31.4	31.7	31.8	30.3	26.4	20.8	18.5		
		MCWB	14.2	15.8	16.5	18.4	21.4	24.1	24.8	24.5	23.0	19.8	16.4	16.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.6	20.6	20.4	22.9	24.4	26.2	27.0	27.3	25.8	24.5	21.6	21.8		
		MCDB	21.5	23.5	24.1	26.9	28.9	31.2	31.4	32.5	30.0	27.9	23.7	23.7		
	2%	WB	18.3	19.6	19.5	21.9	23.5	25.7	26.3	26.5	25.1	23.5	20.3	20.1		
		MCDB	20.3	23.0	23.4	25.6	28.3	30.8	31.4	31.5	29.6	27.2	22.5	22.4		
	5%	WB	16.7	18.6	18.6	21.0	22.9	25.2	25.7	25.8	24.5	22.6	19.0	18.8		
		MCDB	18.2	21.4	22.2	24.5	27.8	30.5	30.7	30.6	29.1	26.0	21.0	20.2		
	10%	WB	14.8	17.0	17.8	20.1	22.4	24.8	25.3	25.3	23.9	21.4	17.6	16.9		
		MCDB	16.9	19.4	21.2	23.4	27.0	29.8	30.2	29.9	28.4	24.5	20.1	18.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.2	12.3	13.1	13.5	12.3	11.5	10.6	10.9	12.2	13.7	13.9	11.8		
		MCDBR	13.0	13.3	14.2	14.3	13.6	13.4	12.1	12.7	13.7	14.6	14.6	12.4		
	5% WB	MCWBR	9.0	7.5	6.8	6.7	6.0	5.0	4.6	4.6	5.1	6.7	8.8	8.5		
		MCWBR	10.1	10.4	10.6	11.0	11.2	11.3	10.6	10.4	10.9	10.8	11.4	10.1		
(40) Clear-Sky Solar Irradiance	taub		0.328	0.329	0.361	0.395	0.427	0.460	0.509	0.508	0.434	0.370	0.344	0.324		
	taud		2.507	2.509	2.429	2.331	2.283	2.224	2.117	2.116	2.307	2.455	2.475	2.534		
	Ebn at Noon		893	929	918	895	865	832	790	785	834	870	866	879		
	Edh at Noon		87	95	111	127	135	142	158	156	124	99	89	80		
(44) All-Sky Solar Radiation	RadAvg		2.74	3.26	4.40	5.52	6.08	6.28	5.99	5.59	5.00	4.18	3.11	2.37		
	RadStd		0.24	0.46	0.36	0.44	0.39	0.47	0.32	0.39	0.36	0.55	0.30	0.24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+116	+71

Nomenclature: See separate page